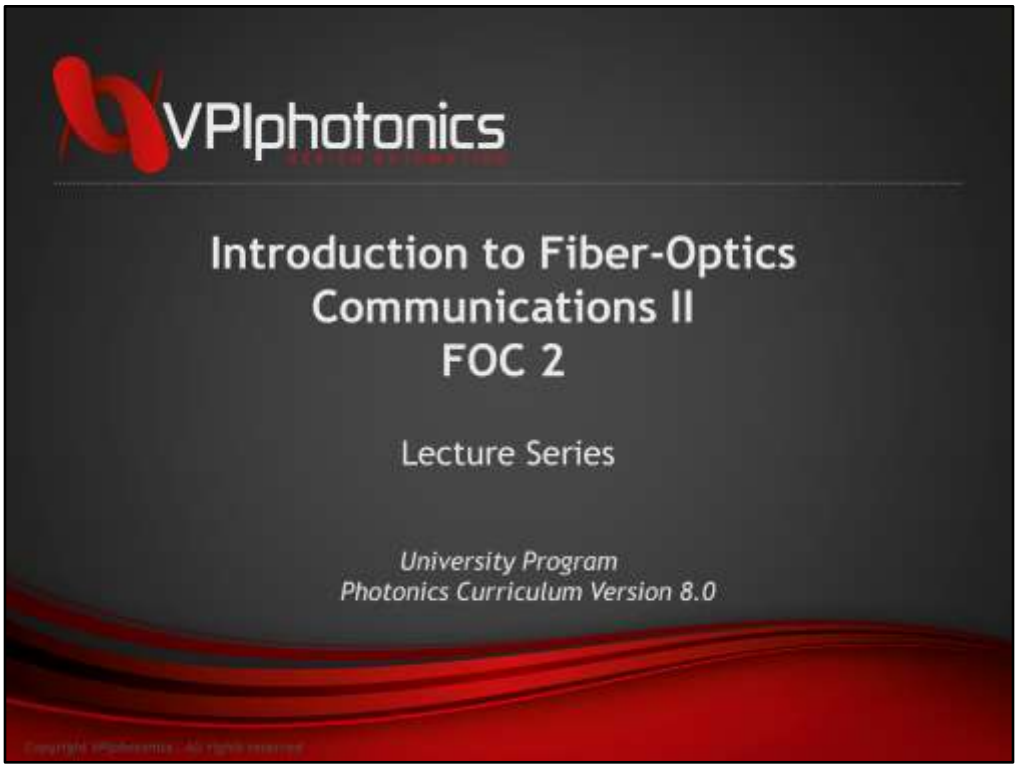

VPIphotonics
Introduction to Fiber-Optics Communications II
FOC 2
Lecture Series
University Program
Photonics Curriculum Version 8.0

Module Prerequisites

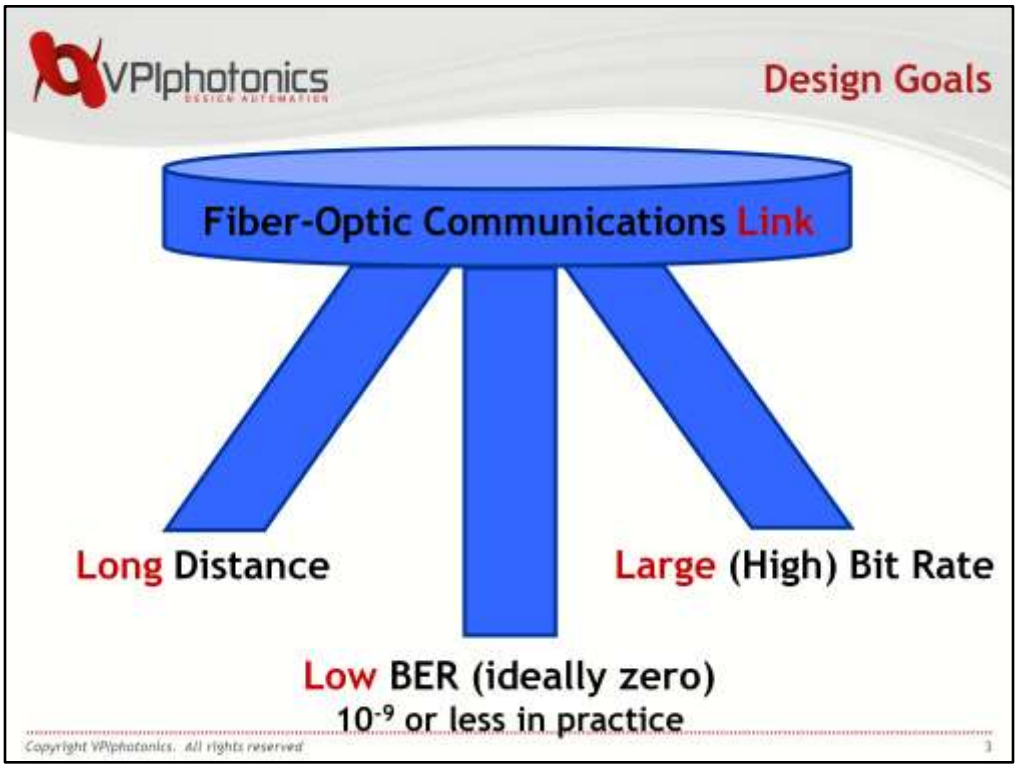
- Introduction to Fiber-Optic Communications I
- Should have worked through the *Simulation Guide* of *VPItransmissionMaker*/*VPIcomponentMaker* before starting this unit (to understand how to handle the software).

Module Objectives

Fiber-Optic Communication Systems

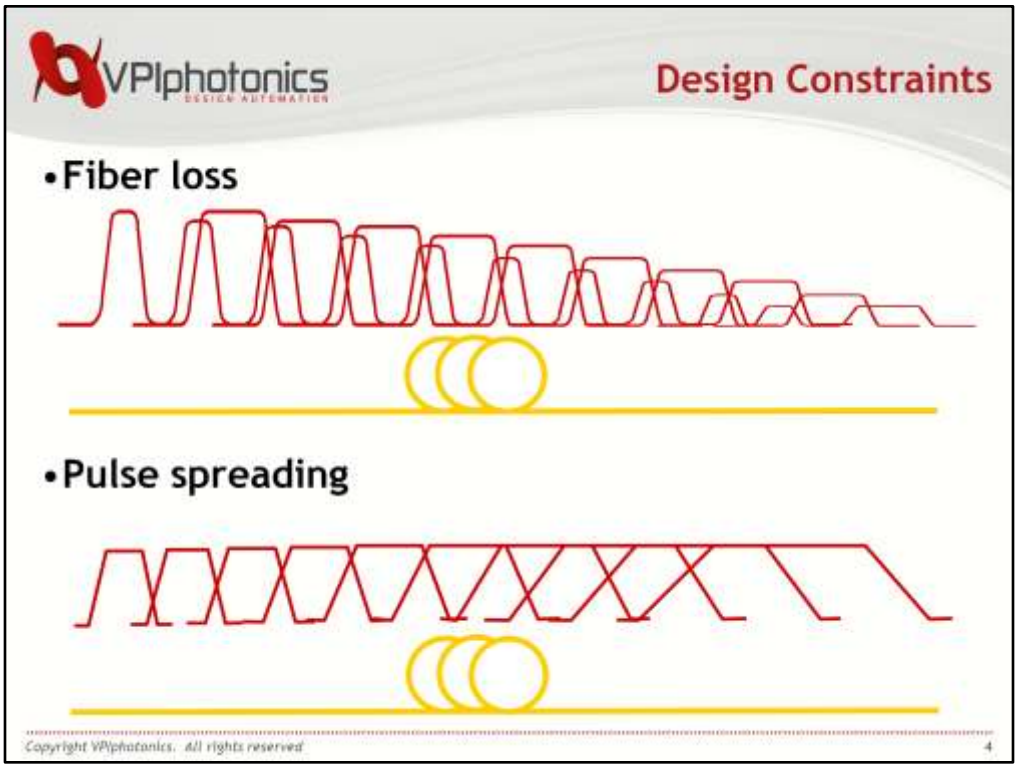
- Design Issues
- Design Constraints
- Link Power Budget

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Design Goals

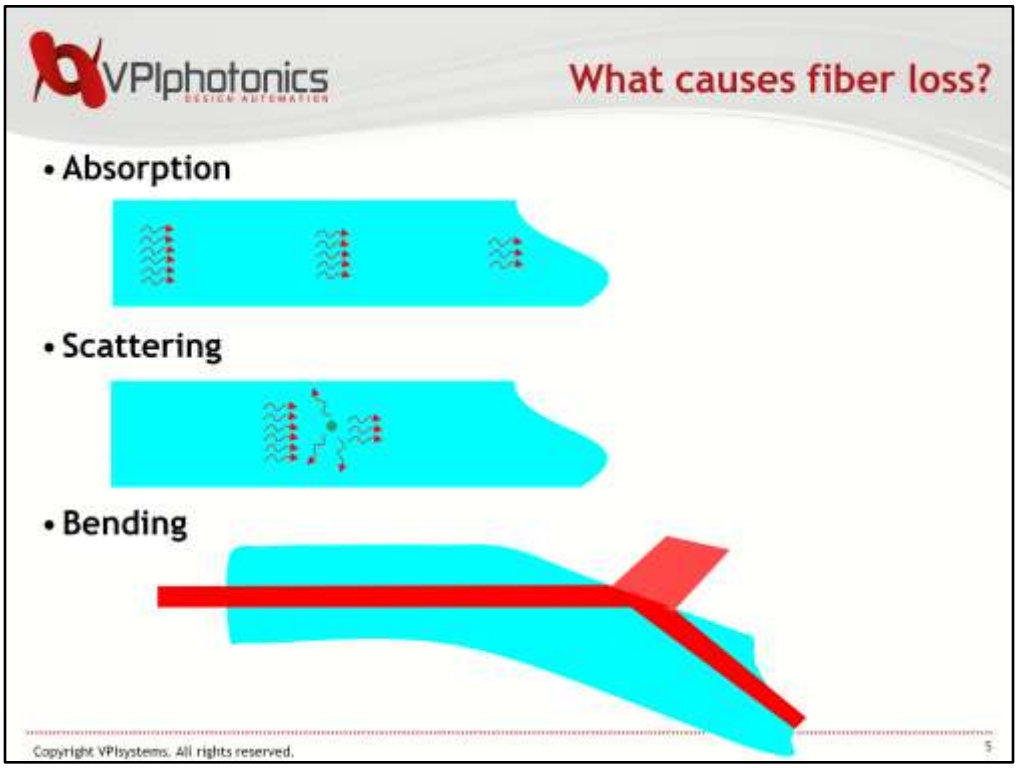
Usually, the bit error rate (BER) of a link or a system is predefined. Nominally, BER is 10^{-9}, or less, for practical systems. Thus, the main design goals of point-to-point fiber-optic communication links are the transmission distance and the transmission bit rate.
Strive for Low BER, Long Distance, Large Bit Rate for the Link.



Design Constraints

What are the constraints?

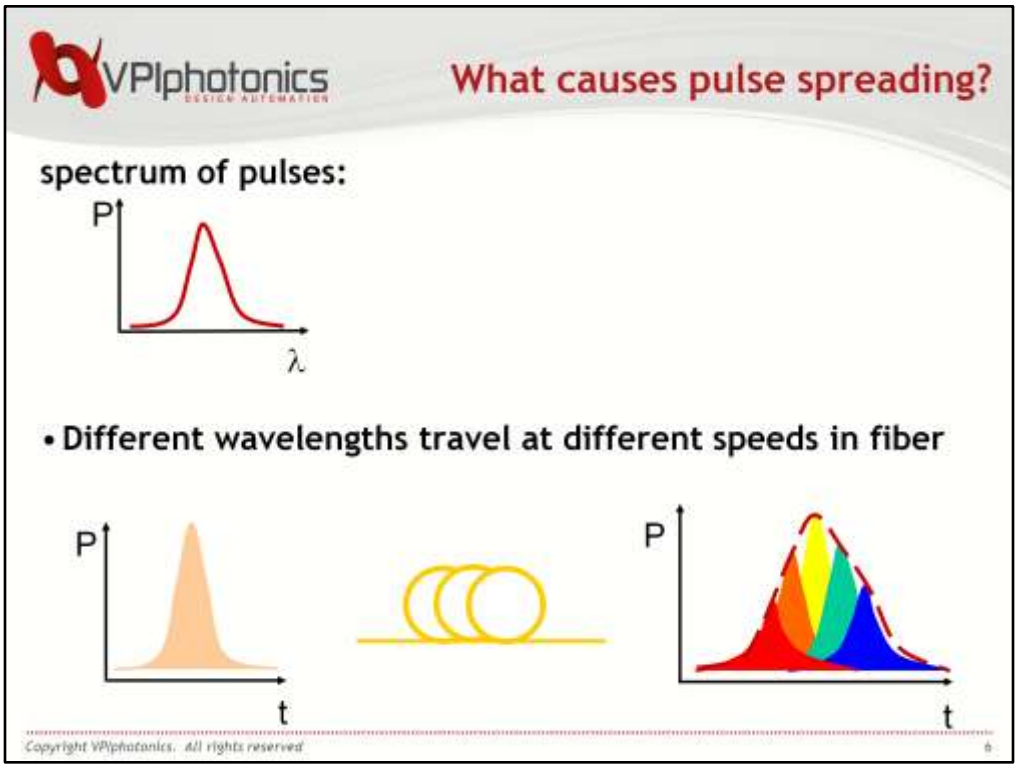
Fiber loss, Pulse spreading (due to fiber dispersion, and chirped transmitting pulses)



Design Constraints

What causes fiber loss?

Various physical effects – intrinsic absorption by atoms and molecules in the glass, extrinsic absorption by impurities in the glass (water molecules, metals), Rayleigh scattering, bending losses etc. More details in another module.

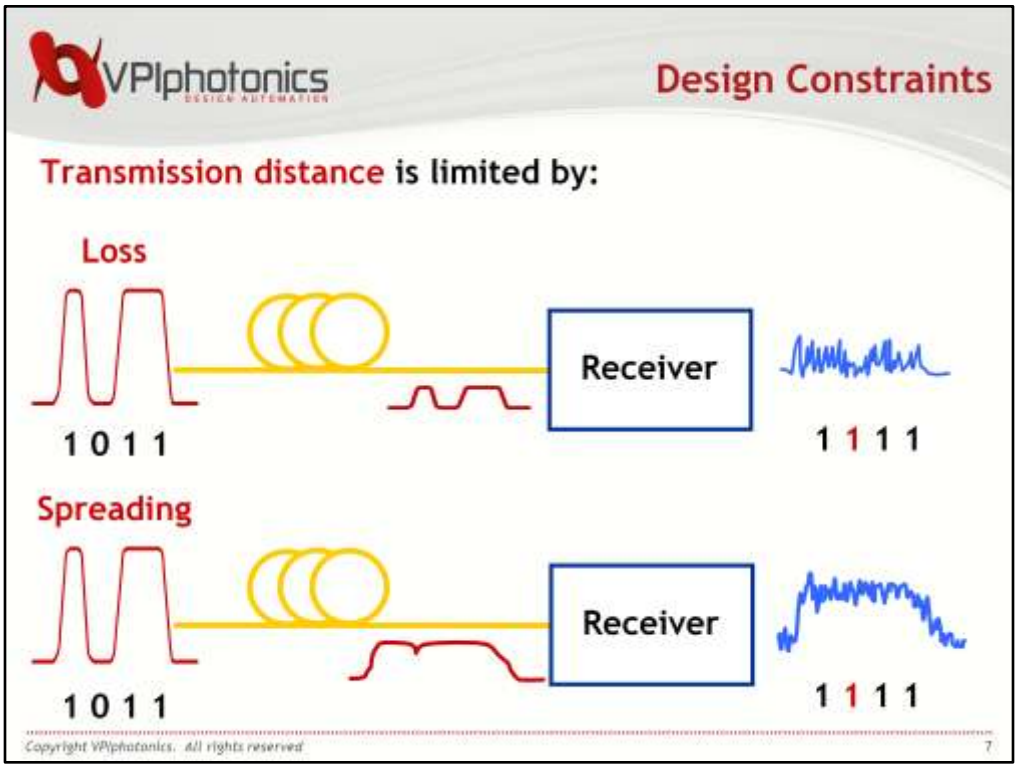


Design Constraints

What causes pulse spreading?

Group velocity dispersion (GVD) in the fibre – different wavelength components of that make up a pulse's spectrum travel at different velocities in glass. Since all pulses (laser source) have a finite linewidth, (spectrum with a finite width), they will experience GVD and spread out in time (I.e. the width of the pulses in the time domain increases with the distance propagated in the fiber). This condition is exacerbated if the pulse spectrum is broadened due to laser chirp.

What is chirp? Chirp is the change in the instantaneous frequency (of the carrier wave) across the pulse. The effect of chirp is to broaden the spectrum of the pulses (laser source). The broadened spectrum exacerbates pulse spreading due to GVD.



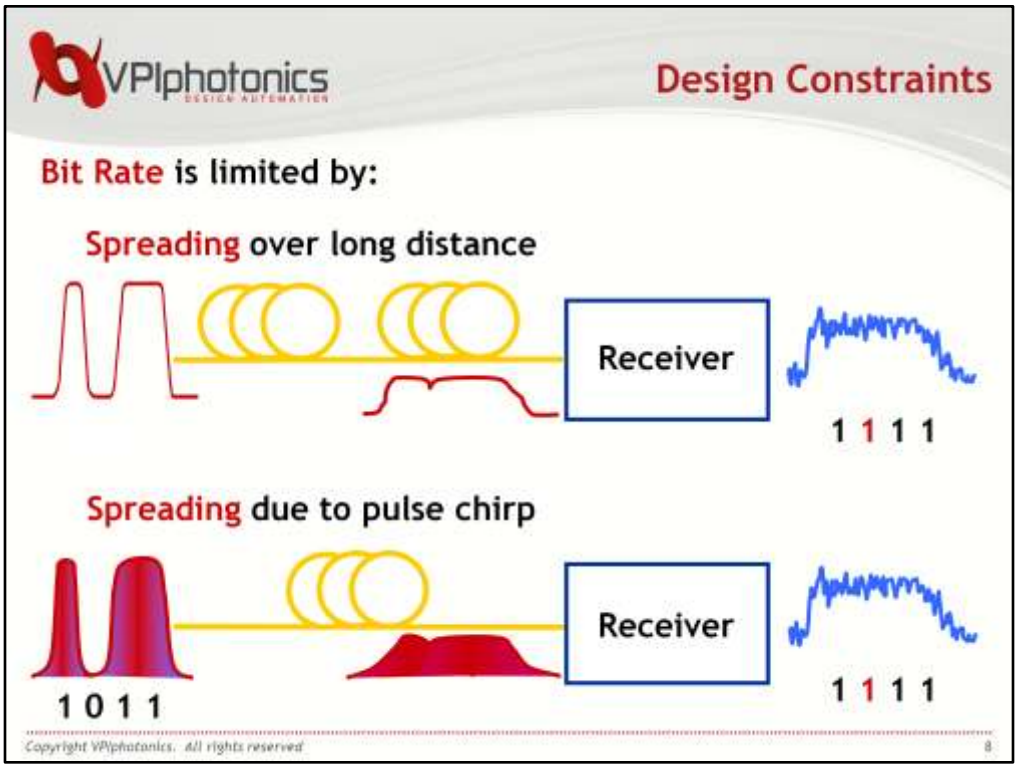
Design Constraints

Distance is limited by: Fiber loss (receiver sensitivity, when thermal noise makes bits indistinguishable)

Pulse spreading due to fiber dispersion, and chirped transmitting pulses

(spreading pulses overlap, making the bits indistinguishable)

(Since the BER performance of the system depends on the average received optical power at the receiver, the transmission distance of the system is also limited by the transmitter's average optical output power. More powerful transmitters can produce pulse streams with higher average power, which enable them to travel further along a fiber before their power level falls below the sensitivity threshold of the system's receiver.)

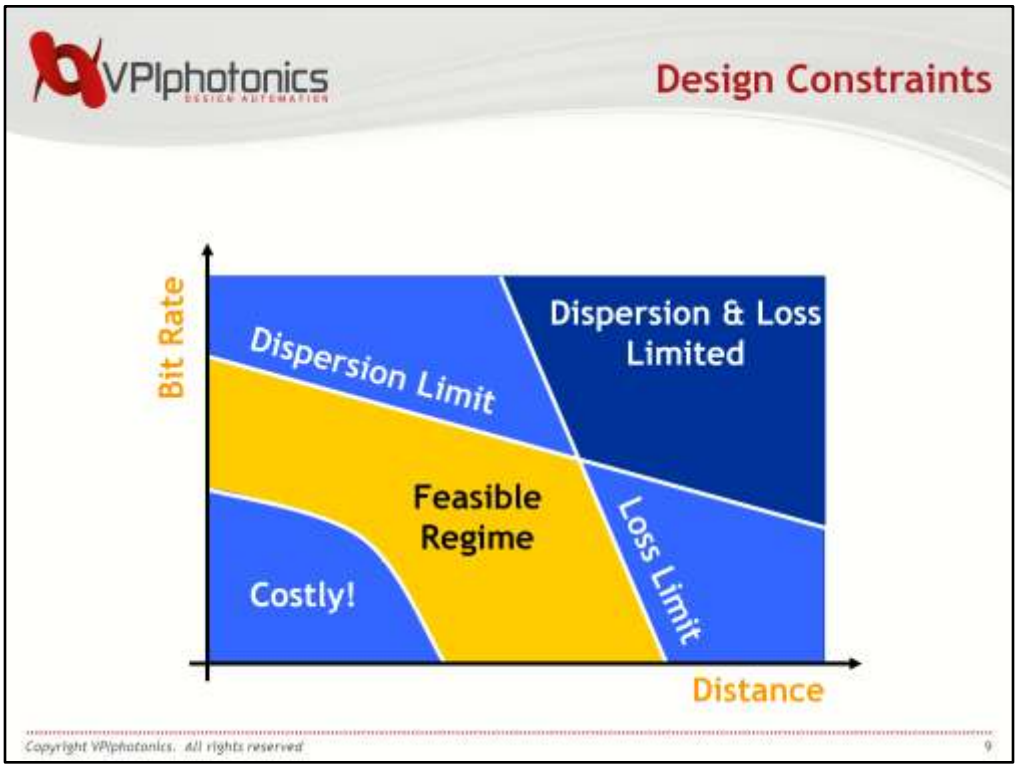


Design Constraints

Bit-Rate is limited by: Pulse spreading due to dispersion over the distance of transmission
Additional pulse spreading due to pulse chirp (broadened spectrum)

The higher the bit rate, the closer the spacing between bits. Over a given (fixed) transmission distance, there is a limit to the attainable transmission bit rate. Limit is reached at a certain bit-rate, where the pulses arrive at the receiver overlapped (making the bits indistinguishable from each other).

(Bit rate is, to a certain extent, also limited by the average transmitter output power. The average power of a higher bit rate pulse stream is higher. Therefore, for a transmitter with a given average output power, the peak power level of the “1” bits will be lower in a higher bit rate pulse stream, than that of a lower bit rate pulse stream. Since the BER is determined by the average received optical power at the receiver, the higher bit rate pulse stream suffers from having a lower averaged received optical power at the receiver.)



Design Constraints: Graphical Representation

For a given bit error rate (BER, nominally 10^{-9} or less for practical systems), the main design goals for point-to-point fiber-optic communication links are the transmission distance (horizontal axis) and the transmission bit rate (vertical axis).

Distance is limited by: Fiber loss (Loss limit)
Pulse spreading due to fiber dispersion, and chirped transmitting pulses

Bit-Rate is limited by: Pulse spreading due to dispersion and pulse chirp (Dispersion limit)

These limits can be represented graphically as shown above.

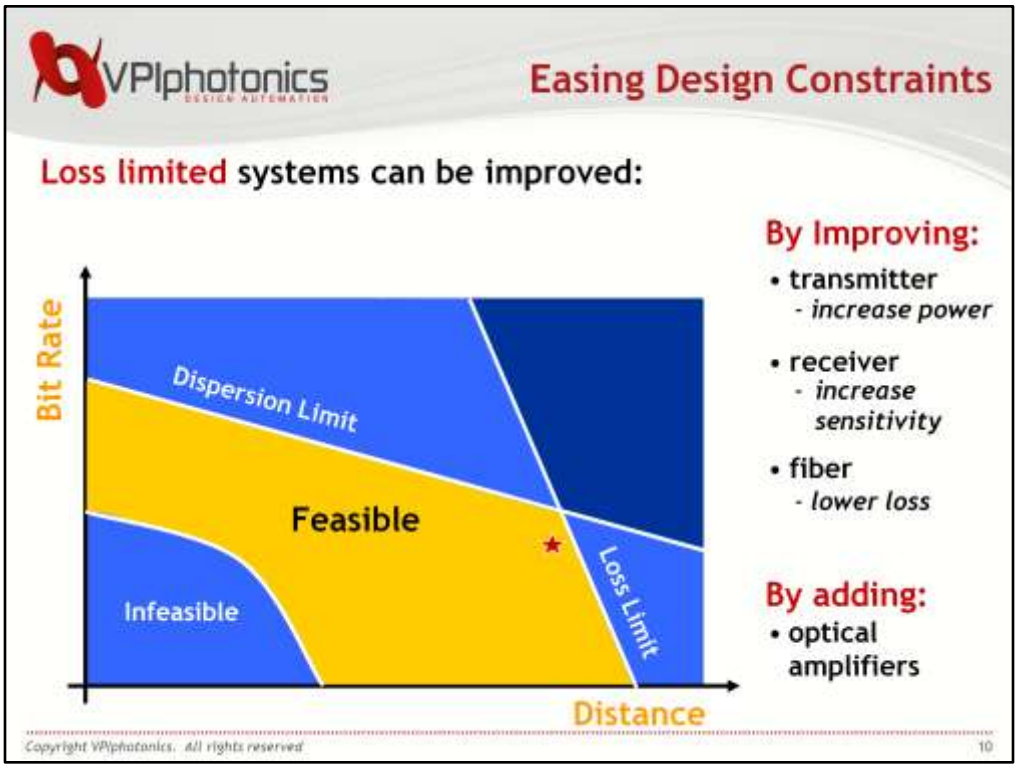
1) The loss limit line delineates the regions where the BER is acceptable (< 10^{-9}) or not (> 10^{-9})

2) Similarly, the dispersion limit line delineates where the BER is acceptable (< 10^{-9}) or not (> 10^{-9})

3) The dark blue region is where the system BER is unacceptable due to both dispersion and loss.

4) The yellow region is where the system is feasible, performance-wise and… <pause for dramatic effect> cost-wise.

5) The lower left-hand corner is where the system is infeasible, cost-wise!



Easing Design Constraints: Loss limited systems

Red star signifies a system that is loss limited. One must either reduce the transmission distance in order to make the system feasible, or one can improve the specification of various parts of the system in order to push back the limit.

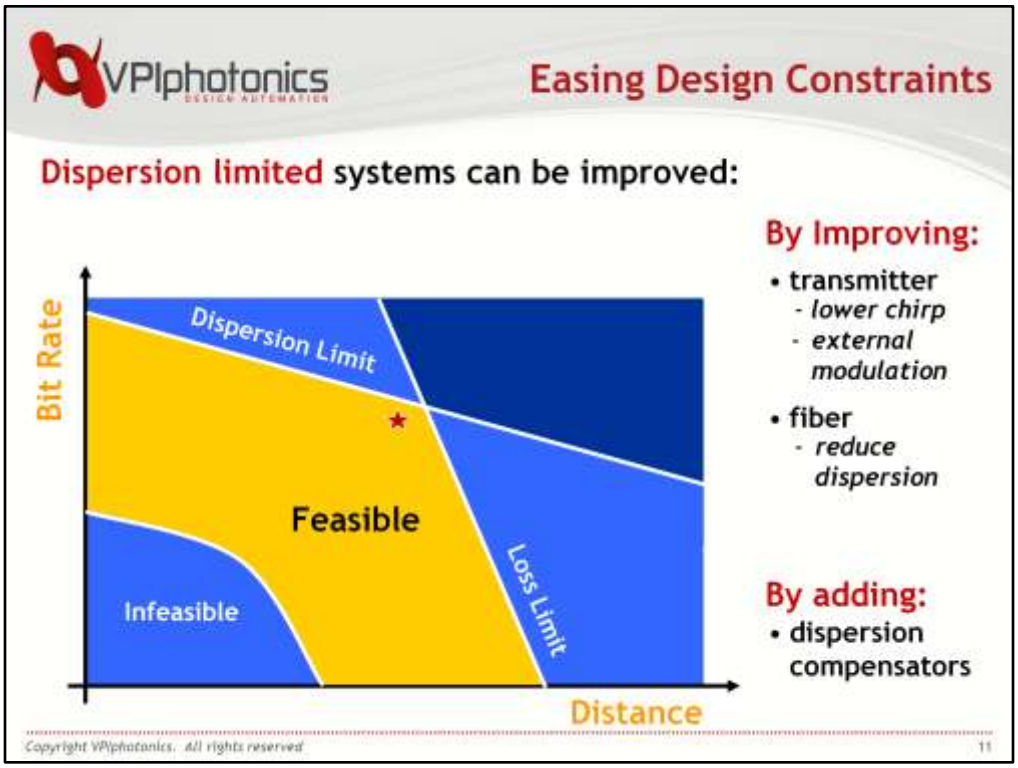
By improving the specifications of:

1) the transmitter (increasing its output power)
2) the fiber (using fibers with lower loss, but note that this is impractical!)
3) the receiver (increasing its sensitivity)

Or

4) Optical amplifiers can be used to extend the transmission distance.

Thus, the loss limit constraint is pushed back, and the system is now feasible.



Easing Design Constraints: Dispersion limited systems

Red star signifies a system that is dispersion limited. One must either reduce the transmission in order to make the system feasible, or one can improve the specification of various parts of the system in order to push back the limit.

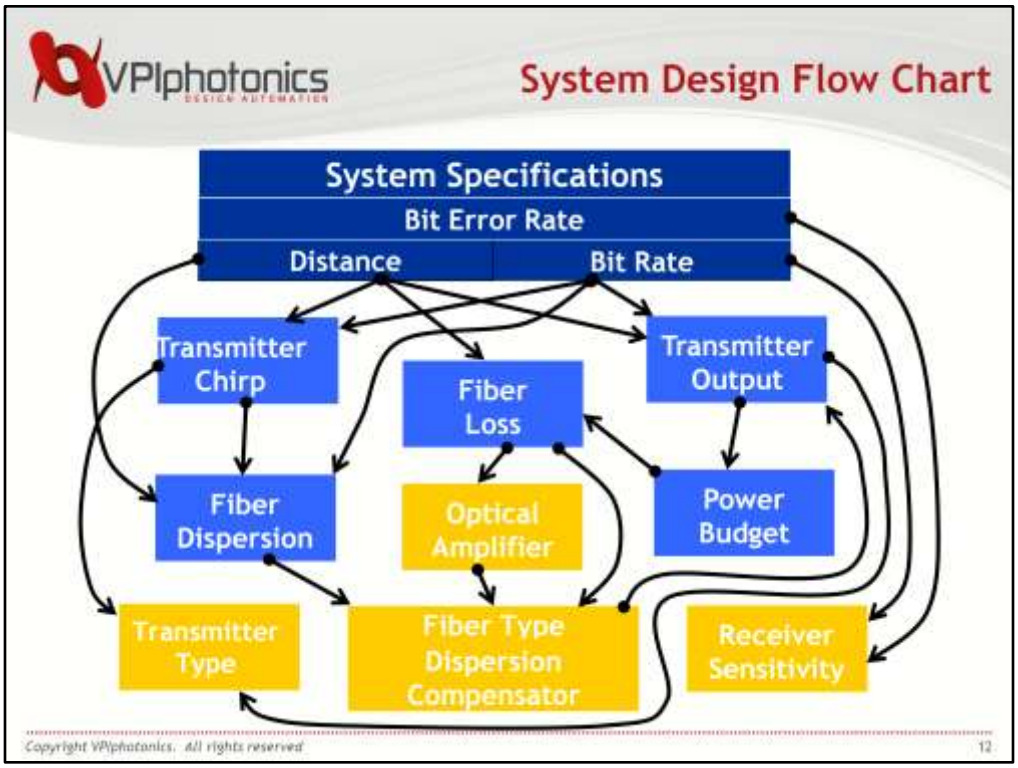
By improving the specifications of:

1) the transmitter (better laser that produces less chirp, or external modulation)
2) the fiber (using fibers with lower dispersion: this may be impractical, and can also enhance the effects of fiber nonlinearity! Such lower dispersion fibers have actually been deployed in Japan.)

Or

3) Dispersion compensators can be used to compensated for the effects of fiber dispersion – this will be covered in more detail in another module.

Thus, the loss limit constraint is pushed back, and the system is now feasible



System Design Flow Chart

1) Start with system specifications

2) The system BER is usually specified at 10^{-9} (or lower), thus the next two design goals are:

(a) Transmission distance and (b) transmission bit rate.

3) Distance is constrained by:

(a) fiber loss, (b) transmitter output power, (c) fiber dispersion and

(d) transmitter chirp, which interacts with fiber dispersion to exacerbate pulse spreading

4) Bit rate is constrained by:

(a) fiber dispersion, (b) transmitter chirp, (c) transmitter output power

5) Constraints can be eased by improving the following parts of the system:

(a) Transmitter – (I) higher output power, (II) less chirp or almost no chirp with external modulation

(b) Receiver – increasing sensitivity to achieved (I) specified BER (II) desired bit rate

(c) Fiber – using fibers with (I) lower loss, (II) lower dispersion,

the choices of which will have an impact on (III) required transmitter output power.

6) Usually, there is not much we can do about the fiber, and there are significant ramifications (which will be discussed in another module on fibers) to using non-standard transmission fibers (effects of nonlinearities). Therefore, the improvement of the fiber part of the system is achieved by:

(a) using optical amplifiers to compensate for fiber loss, and

(b) using dispersion compensators to compensate for the dispersion

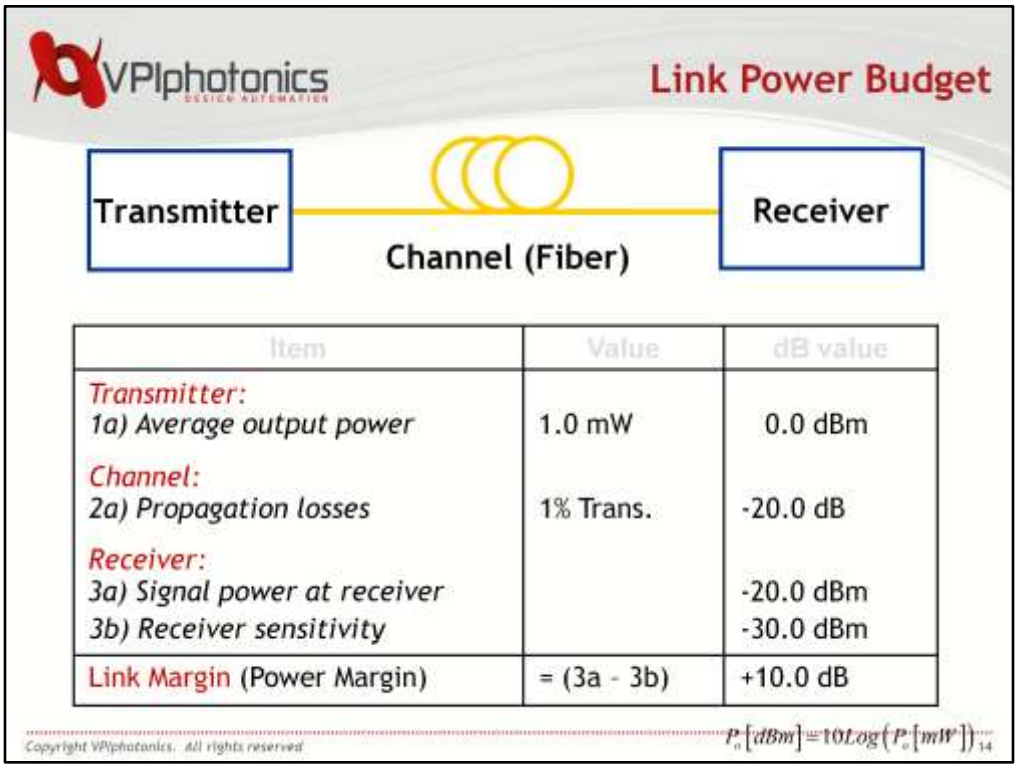
Because a system's BER is dependent on the averaged received optical power at the receiver, a straight forward method of ascertaining if the performance of a particular system is acceptable is to perform a

Link Power Budget: dBm

$$P_o[dBm] = 10Log(P_o[mW])$$

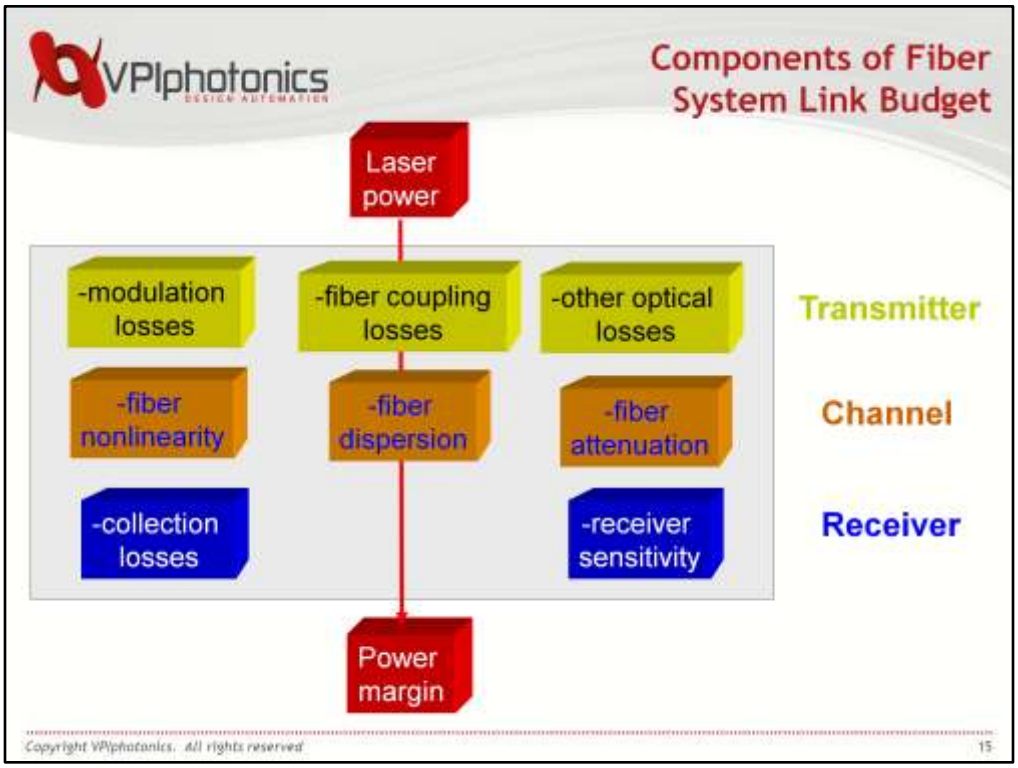
- 1 μW
- 10 μW
- 100 μW
- 1mW
- 2 mW
- 4 mW
- 10 mw
- 80 mW
- 100 mW

- -30 dBm
- -20 dBm
- -10 dBm
- 0 dBm
- +3 dBm
- +6 dBm
- +10 dBm
- +19 dBm
- +20 dBm



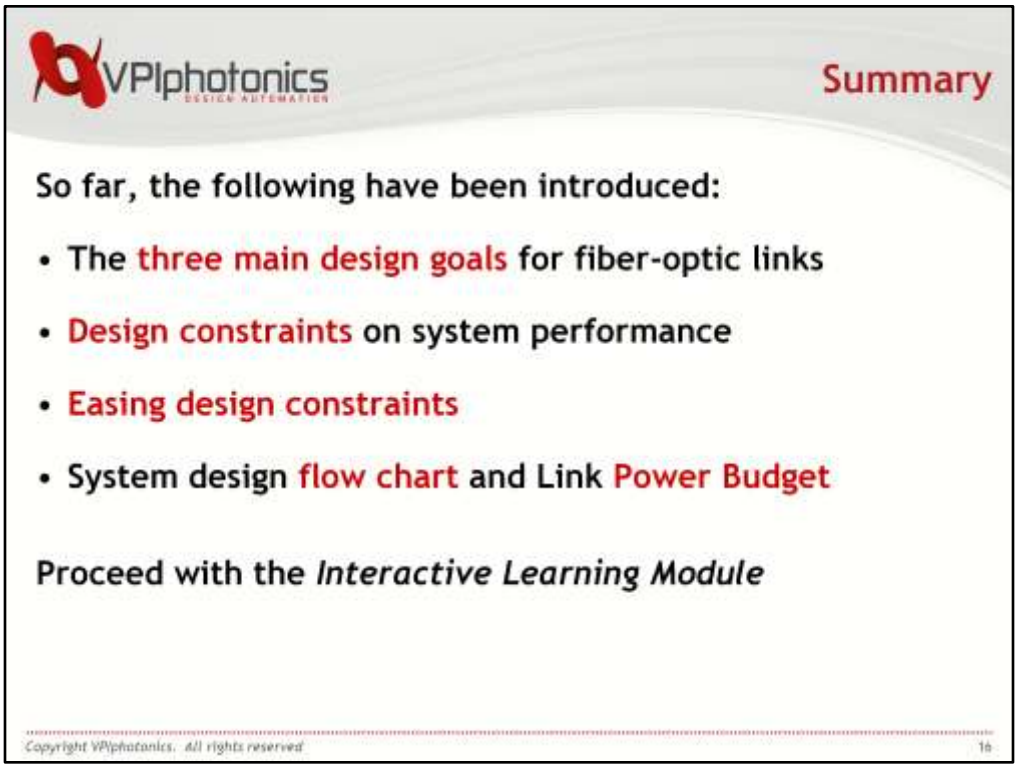
Link Power Budget

The overall system level performance of an optical communication link is easily quantified in terms of a system link budget.



Link Power Budget

The various parts of the Fiber System Link Budget can be broken down further to various components, in order to gain a clearer picture of which part of the system is limiting the performance.



Summary of the key points covered in the lecture part of this module.

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